

Lecture 1: slide-by-slide teaching script

Introduction and Review of Probability

Scope. This guide follows `Lecture1v2.tex` from the title slide through *Correlation*, then stops before *Random Sampling*. The slide numbers below count the title page as slide 1. Use the slide title as the primary locator if the compiled deck displays a different page number.

How to use this guide. For every slide, first use the “Say” line, then do the board or screen action, then ask the short question. Do not read the slide verbatim. The worked examples are deliberately attached to a particular slide so that the calculation has a clear purpose.

Class route

Minutes	Slides and purpose
0–5	1–2: establish the course question and assessment context.
5–18	3–6: move from economic questions to causal questions and the course map.
18–35	7–9: random variables, density, and cumulative probability.
35–50	10–12: moments, variance calculation, and the earnings example.
50–54	Short pause; ask students to write one unresolved question.
54–80	13–16: joint probabilities, conditioning, and iterated expectations.
80–94	17–20: independence, covariance, and correlation.
94–100	Exit check and preview of Random Sampling.

Slide-by-slide script

Slide 1 | Lecture 1 v2: Introduction and Review of Probability

Say Welcome students and give the class its organizing question: how can data help us learn about economic relationships when outcomes are uncertain? Explain that probability is the language used to describe that uncertainty.

Do Do not start with notation. Point to the title, state that today’s class has two halves, and tell students that the second half will connect directly to regression.

Example Give a one-sentence preview: “If education and earnings move together, what would convince us that education caused the change?” Do not solve it yet.

Transition Move from the broad question to what the course assesses and how the course is organized.

Slide 2 | About this course

Say Briefly explain the assessment weights and set the expectation that participation means trying the calculations, not merely answering correctly.

Do Spend at most two minutes here. Point out where lecture material and readings will be available.

Ask “Which part of the assessment rewards steady work throughout the term?” Use the answer to frame problem sets and participation.

Transition The course is about applying statistical tools to economic questions, so now introduce the type of question those tools are meant to answer.

Slide 3 | What’s Econometrics about?

Say Econometrics combines an economic question, data, and statistical methods. The central output is a quantified relationship, but the meaning of that relationship depends on the question being asked.

Example Use education and earnings. Ask: “Do graduates earn more?” Then change the wording to: “What would happen to one person’s earnings if that person received one more year of education?” Emphasize that these are different questions.

Board / screen: Write: *association* = what differs across observed people; *causal effect* = change under an intervention.

Ask Take two student answers about why education may be related to earnings: ability, family background, motivation, or field of study. Keep the list visible for slide 4.

Transition The difference between those two questions is the difference between describing data and identifying a causal effect.

Slide 4 | Inferring Causal Effects (and more)

Say A randomized experiment makes treatment assignment unrelated to pre-existing characteristics, at least in expectation. Observational data do not automatically have that property.

Example Use a class-size experiment: randomly assign some classes to 15 students and others to 30, then compare achievement. Ask what the randomization is buying us.

Comment Say explicitly that randomization does not guarantee identical groups in one realized sample. It removes systematic selection on average; chance imbalance can remain.

Ask “Why can we not randomly assign people to different years of education?” Accept ethical, practical, and cost concerns, then point out that econometrics needs assumptions and designs for observational data.

Transition Before learning those designs, students need a common language for uncertainty and association.

Slide 5 | A sentence you are going to hear a lot

Say Read the sentence as a warning about interpretation: correlation is evidence that two variables move together, not by itself evidence that one causes the other.

Example Return to education and earnings. More education may correlate with earnings because education changes earnings, because ability affects both, or because of another common factor.

Ask Have students give one example from daily life where two variables are correlated but neither clearly causes the other. Keep one example, such as umbrellas and rain, and move on.

Transition The course will revisit this distinction in every block. The outline shows where the first tools fit.

Slide 6 | Course Outline

Say Give the map in three sentences: first probability, statistics, and simple regression; then multivariate regression and inference; finally identification threats, instruments, panel data, and binary choice.

Do Point out where the two midterms sit, but do not lecture on the dates. Students need the structure, not a reading of every bullet.

Example Tell them today’s probability tools will reappear as conditional means, covariance, sampling distributions, and OLS slopes.

Transition Start the review with the smallest unit: a random variable and the outcomes it summarizes.

Slide 7 | Some Revision

Say Separate four objects: outcome, sample space, event, and random variable. A random variable is a numerical mapping from uncertain outcomes; it is not itself the outcome.

Worked example Use one fair coin toss. Outcomes are $\{H, T\}$. The sample space is $\{H, T\}$. The event “heads” is $\{H\}$. Define $Y = 1$ for heads and $Y = 0$ for tails, so $P(Y = 1) = P(Y = 0) = .5$.

Board / screen: Ask students what changes if heads is coded as 10 and tails as 0. The probabilities of the outcomes do not change, but the numerical mean and variance of the coded variable do.

Ask “What must the probabilities of all mutually exclusive values add up to?” Answer: one. Then distinguish a discrete probability mass function from a continuous density.

Transition For a continuous variable, individual points have probability zero, so we use areas and cumulative probabilities.

Slide 8 | P.d.f. of a continuous random variable

Say A density is not a probability at a point. Probability is the area under the density over an interval, and the total area is one.

Worked example Use a waiting time W uniformly distributed between 0 and 10 minutes. Draw a rectangle of height 0.1. Then calculate $P(2 < W \leq 5)$ as the rectangle area: $(5 - 2)(0.1) = 0.3$.

Board / screen: $f_W(w) = 0.1$ for $0 \leq w \leq 10$; shade the interval from 2 to 5. Label the shaded area 0.30.

Ask “Can a density be greater than one?” Yes. A uniform variable on $[0, .5]$ has density 2; its area is still one.

Transition The density gives interval areas. The next slide turns those areas into a cumulative function that is easier to evaluate.

Slide 9 | C.d.f. of a continuous random variable

Say The c.d.f. at w is $F_W(w) = P(W \leq w)$. It must be nondecreasing, start at zero, and approach one.

Worked example Continue the same waiting-time example: $F_W(w) = w/10$ for $0 \leq w \leq 10$. Calculate $F_W(5) = .5$, then recover the previous interval probability as $F_W(5) - F_W(2) = .5 - .2 = .3$.

Board / screen: Draw the c.d.f. as a straight line from $(0, 0)$ to $(10, 1)$. Mark $w = 2$ and $w = 5$ and show the vertical difference in c.d.f. values.

Ask “What is $P(W = 3)$?” For a continuous variable it is zero: a single point has no area.

Transition Now move from probabilities of values and intervals to numerical summaries of a distribution.

Slide 10 | Statistical Moments

Say The mean locates a distribution, the variance measures squared dispersion, and the standard deviation returns dispersion to the original units. Show the definition of variance on the slide and tell students the next slide gives the practical shortcut $\text{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$.

Do Point to each moment and give the interpretation before doing algebra: mean is center, variance is dispersion, skewness is asymmetry, kurtosis is tail mass.

Comment Variance has squared units, so standard deviation is often easier to interpret. Keep this slide conceptual; the numerical work starts immediately on Slide 11.

Transition Before moving to skewness and kurtosis, pause for one full calculation slide so students see how variance can be computed in practice.

Slide 11 | Calculating variance two ways

Say This slide is a worked calculation, not a new definition. The point is that the two variance formulas produce the same number.

Do Start on the left. Compute the mean first: $\mathbb{E}(Y) = 2.25$. Then plug each value into the deviation formula and get $\text{Var}(Y) = 1.1875$.

Shortcut Move to the right side and compute $\mathbb{E}(Y^2) = 6.25$. Then show $\text{Var}(Y) = \mathbb{E}(Y^2) - [\mathbb{E}(Y)]^2 = 6.25 - 2.25^2 = 1.1875$. Circle both final values.

Linear combination preview Use the last formula only as a preview. Say: when a variable is a linear combination, the variance is not only the sum of the scaled variances; there is an extra average product term, $\mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$, that captures whether the two parts tend to move together. Do not name this term yet.

Comment Leave the linear-combination formula as a preview only. Explain that the first two terms are the scaled variances and the final term captures whether the two components move together. Do not give it a name or work through a separate example yet.

Ask “Why do we square a and b in the formula?” Because multiplying a variable by a multiplies its deviations by a , and variance squares deviations.

Transition Now return to the shape summaries: skewness and kurtosis describe features beyond location and dispersion.

Slide 12 | Examples

Say Use the figures to connect signs to shapes: positive skew means a longer right tail, negative skew a longer left tail, and high kurtosis means more mass in the tails than a normal reference.

Worked example Give a simple wage story: most workers earn around 2, but a small number earn 10. That creates a long right tail and positive skewness. Ask what a small number of unusually low wages would do instead.

Comment Do not claim that zero skewness guarantees symmetry. Also say that kurtosis is sensitive to extreme values and that the exact benchmark depends on the convention used.

Fast check If every wage increases by 100 euros, the mean increases by 100 euros but variance and standard deviation do not change. If every wage doubles, the standard deviation doubles and variance becomes four times as large.

Transition One variable is no longer enough when we ask whether sunny days and ice-cream purchases occur together. Build their joint distribution.

Slide 13 | Joint Probability Distribution

Say Introduce $X = 1$ for a sunny day and $Y = 1$ for an ice-cream purchase. The internal cells describe joint events; the margins describe one variable at a time.

Worked example Use the slide’s table and ask students to calculate $P(Y = 1)$ by adding the row: $.05 + .55 = .60$. Then calculate $P(X = 1)$ by adding the column: $.25 + .55 = .80$.

	$X = 0$	$X = 1$	Total
$Y = 0$	.15	.25	.40
$Y = 1$	.05	.55	.60
Total	.20	.80	1

Comment Say “joint probability uses the full population of days.” This prevents the common mistake of treating .55 as a conditional probability.

Ask “What is $P(X = 1, Y = 1)$?” Answer: .55, the bottom-right cell.

Additional example Replace the weather story for two minutes with a study-workshop example. Let $X = 1$ mean a student attended a review workshop and $Y = 1$ mean the student passed. Use the table below and ask students to identify one joint probability and both marginal probabilities.

	$X = 0$	$X = 1$	Total
$Y = 0$	.30	.10	.40
$Y = 1$	.20	.40	.60
Total	.50	.50	1

Answers Here $P(X = 1, Y = 1) = .40$, $P(Y = 1) = .60$, and $P(X = 1) = .50$. Say explicitly that the table describes an association between workshop attendance and passing; it does not yet prove that the workshop caused the pass.

Transition The next slide uses this exact table to calculate marginals, joint probabilities, and conditionals slowly.

Slide 14 | Working with the joint distribution

Say Use this slide to slow down. The same table can answer three different kinds of probability questions: marginal, joint, and conditional.

Marginals Point to the totals. Calculate $P(Y = 1) = .05 + .55 = .60$ and $P(X = 1) = .25 + .55 = .80$. Say: these use whole rows or columns.

Joint Point to the bottom-right internal cell: $P(X = 1, Y = 1) = .55$. Say: this is not conditional yet; the denominator is the whole population of days.

Conditionals Now change the denominator. For $P(Y = 1 | X = 1)$, use the $X = 1$ column as the reference group: $.55/.80 = .6875$. For $P(X = 1 | Y = 1)$, use the $Y = 1$ row: $.55/.60 \simeq .9167$.

Ask “Why do both conditional probabilities use the same numerator but different denominators?” Because they condition on different reference groups.

Transition Conditional probabilities become conditional expectations when the variable has numerical values. That is the bridge to the next slide.

Slide 15 | Conditional and Iterated Expectations

Say When we condition on $X = 1$, we restrict attention to sunny days. The denominator is therefore $P(X = 1) = .80$, not one.

Worked example Calculate

$$P(Y = 1 | X = 1) = \frac{.55}{.80} = .6875, \quad P(Y = 1 | X = 0) = \frac{.05}{.20} = .25.$$

For binary Y , the conditional mean equals the conditional probability of one, so $\mathbb{E}(Y | X = 1) = .6875$.

Lead-in Do not do the full law of iterated expectations calculation here anymore; it now has its own slide. Use this slide to define the objects and identify which probabilities from Slide 14 will be used.

Additional example Use a numeric outcome to show why conditional expectation is not limited to probabilities. Suppose hourly wage is 20 with probability .70 and 12 with probability .30 among degree holders. Then $\mathbb{E}(W | D = 1) = .70(20) + .30(12) = 17.60$. If the corresponding non-degree probabilities are .20 and .80, then $\mathbb{E}(W | D = 0) = .20(20) + .80(12) = 13.60$.

Reverse conditioning Return to the sunny-day table and calculate $P(X = 1 | Y = 1) = .55/.60 = 11/12 \simeq .9167$. Contrast it with $P(Y = 1 | X = 1) = .55/.80 = .6875$: the numerator is the same, but the reference population and denominator change.

Ask “Why is averaging .6875 and .25 with equal weights wrong?” Because the two conditioning groups do not have equal probabilities.

Transition Now calculate the law explicitly: first average inside each X group, then average those group averages.

Slide 16 | Calculating the law of iterated expectations

Say Because Y is binary, the conditional expectation equals the probability that $Y = 1$ inside the conditioning group.

Step 1 Write the two conditional means:

$$\mathbb{E}(Y \mid X = 1) = .55/.80 = .6875, \quad \mathbb{E}(Y \mid X = 0) = .05/.20 = .25.$$

Step 2 Apply the law using the correct group weights:

$$\mathbb{E}[\mathbb{E}(Y \mid X)] = .6875(.80) + .25(.20) = .55 + .05 = .60.$$

Verification Calculate the same value from the marginal distribution: $\mathbb{E}(Y) = 1P(Y = 1) + 0P(Y = 0) = .60$. Circle the two final .60 values and say: average within groups, then average across groups using group probabilities.

Warning Tell students not to average .6875 and .25 with equal weights. Sunny days are four times as common as non-sunny days, so the weights are .80 and .20.

Transition If conditioning did not change the probability of ice cream, the variables would be independent.

Slide 17 | Independence

Say Independence means knowing X gives no information about Y : $P(Y = y \mid X = x) = P(Y = y)$ for every relevant pair. It is a statement about the distribution, not a claim that one variable causes the other. An equivalent factorization is $P(X = x, Y = y) = P(X = x)P(Y = y)$.

Worked example Use two dice as the clean positive example: the value of die 1 does not change the distribution of die 2, so $P(X = x, Y = y) = P(X = x)P(Y = y)$.

Dice calculation Let $X = 1$ mean the first die shows a six and $Y = 1$ mean the second die is even. Then $P(X = 1) = 1/6$, $P(Y = 1) = 1/2$, and $P(X = 1, Y = 1) = 1/12 = (1/6)(1/2)$. Also $P(Y = 1 \mid X = 1) = 1/2$. Use this to make the factorization tangible.

Simple expectation calculation Tell students that the next slide gives the full joint table for this dice example and shows $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ numerically.

Mathematical proof, discrete case For discrete X and Y , show exactly where independence is used:

$$\begin{aligned} \mathbb{E}(XY) &= \sum_x \sum_y xy P(X = x, Y = y) \\ &= \sum_x \sum_y xy P(X = x)P(Y = y) \quad (\text{independence}) \\ &= \left[\sum_x xP(X = x) \right] \left[\sum_y yP(Y = y) \right] \\ &= \mathbb{E}(X)\mathbb{E}(Y). \end{aligned}$$

Say that the double sum separates into two sums because the joint probability factors.

Dependent example Draw two cards without replacement. Let $X = 1$ mean the first card is an ace and $Y = 1$ mean the second card is an ace. Before seeing the first card, $P(Y = 1) = 4/52 = 1/13$; after observing $X = 1$, $P(Y = 1 \mid X = 1) = 3/51 = 1/17$. Because these differ, the draws are not independent.

Check the class table For sunny days and purchases, independence would require $.8(.6) = .48$ in the bottom-right cell. The table gives .55, so independence fails.

Ask “Does failure of independence prove that sunshine causes purchases?” No. It only says the variables are statistically related; other explanations remain possible.

Transition Use the next slide to make independence visible in a joint table before introducing covariance.

Slide 18 | An independent joint distribution

Say This table looks like the previous joint table, but now every internal cell equals the product of its row and column margins. That is independence.

Joint check Use the bottom-right cell:

$$P(X = 1, Y = 1) = 1/12 = (1/6)(1/2) = P(X = 1)P(Y = 1).$$

Then say the same check works for the other three cells.

Conditional check Show that conditioning does not change the probability:

$$P(Y = 1 | X = 1) = \frac{1/12}{1/6} = 1/2 = P(Y = 1).$$

Expectation check Because X and Y are indicators, $\mathbb{E}(X) = 1/6$ and $\mathbb{E}(Y) = 1/2$. Also $XY = 1$ only in the bottom-right cell, so $\mathbb{E}(XY) = 1/12$. Hence

$$\mathbb{E}(XY) = 1/12 = (1/6)(1/2) = \mathbb{E}(X)\mathbb{E}(Y).$$

Proof reminder Connect to the proof from Slide 17: independence lets the joint probability factor, and that is why the expectation of the product factors.

Transition When independence fails, the difference between $\mathbb{E}(XY)$ and $\mathbb{E}(X)\mathbb{E}(Y)$ becomes useful. Now give that difference its name.

Slide 19 | Covariance

Say Covariance multiplies each variable’s deviation from its mean and averages those products. Same-sign deviations contribute positively; opposite-sign deviations contribute negatively. Show the equivalent shortcut formula:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

Worked example For the binary table, use $XY = 1$ only in the bottom-right cell, so $\mathbb{E}(XY) = .55$. Then

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = .55 - (.8)(.6) = .07.$$

This is positive, matching the table’s pattern.

Difference from correlation Covariance is measured in the product of the units of X and Y , and its magnitude has no fixed upper bound. It changes when a variable is rescaled: if Y is recorded in tens, $\text{Cov}(X, 10Y) = 10 \text{Cov}(X, Y) = .70$. Adding a constant does not change it, so $\text{Cov}(X, Y + c) = \text{Cov}(X, Y)$. The sign is useful, but the raw size is harder to compare across measurements.

Independence implication Now connect Slides 17 and 18 in one line:

$$X \perp Y \Rightarrow \mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y) \Rightarrow \text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = 0.$$

Say that independence implies zero covariance, provided the expectations exist.

Comment The converse is false: zero covariance does not generally imply independence. Do not let the class get stuck interpreting the units of .07.

Optional side example If time allows, let $X \in \{-1, 0, 1\}$ with equal probabilities and $Y = X^2$. Then Y is completely determined by X , but $\text{Cov}(X, Y) = 0$ because $\mathbb{E}(X^3) = 0$.

Transition Because covariance changes with measurement units, normalize it by the two standard deviations.

Slide 20 | Correlation

Say Correlation is standardized covariance and always lies between -1 and 1. It describes the direction and strength of linear association, not a causal effect and not a percentage change.

Worked example Continue the same table. Since $\text{Var}(X) = .8(.2) = .16$ and $\text{Var}(Y) = .6(.4) = .24$,

$$\rho_{XY} = \frac{.07}{\sqrt{.16}\sqrt{.24}} \simeq .3572.$$

Interpret this as a moderate positive association in this illustrative population.

Compare Correlation removes the scale and units by dividing by both standard deviations. It is unit-free, bounded between -1 and 1, and unchanged by positive rescaling or adding constants: $\rho_{X,10Y} = \rho_{XY} = .3572$. Thus covariance keeps the original measurement scale, while correlation makes linear association easier to compare across variables.

Ask “Does .3572 mean that sunshine increases purchases by 35.72%?” No. It is a unit-free linear association measure.

Close the route Point back to the opening sentence: the table describes a relationship, but causal interpretation requires a design or additional assumptions. Tell students that the next section begins with random samples and asks how population quantities can be learned from data.

End-of-class check

1. Explain the difference between $P(Y = 1, X = 1) = .55$ and $P(Y = 1 | X = 1) = .6875$.
2. Why is $\mathbb{E}(Y | X = 1)$ a probability when Y is binary?
3. Does zero covariance imply independence? Give the $Y = X^2$ counterexample if asked.
4. Does random assignment guarantee exact balance in one sample? No; it balances in expectation.

Preparation for the next class: ask students to compute $P(X = 1 | Y = 1) = .55/.60 = 11/12$ and explain why reversing the conditioning changes the denominator. Preview that Random Sampling will distinguish a population mean from a sample mean.

Sources: `Lecture1v2.tex` and `Syllabus_2026_2027.tex`. Waiting-time, earnings, dice, and $Y = X^2$ calculations are instructor-created illustrations. Timing assumes a 100-minute session.